



Seat No : _____

No. of Printed Pages: 2

[12]

SARDAR PATEL UNIVERSITY
BCA Semester-6 Examination – July-2023
US06CBCA23 || DATA COMMUNICATION AND COMPUTER NETWORKS

Date : 25-7-2023

Time : 10.00 a.m. – 01.00 p.m.

Total : 70 Marks

Q-1 **Select the appropriate option for the following questions:** **10**

- 1) _____ have a single communication channel that is shared by all the machines on the network.
A. Point-to-Point network. C. Broadcast network.
B. People network D. Specialized network
- 2) The connection of two or more networks is called an _____.
A. computer network C. internetwork
B. internet D. local Area Network
- 3) A _____ cable is made of glass or plastic and transmits signals in the form of light.
A. Twisted-pair C. Fiber Optical
B. Coaxial cables D. None of these.
- 4) In _____ Transmission is use n wires to send n bits at one time.
A. Serial C. Simplex
B. Parallel D. Half duplex
- 5) A list of the protocols used by a certain system, one protocol per layer, is called a _____.
A. network architecture C. Computer architecture
B. Layers stack D. protocol stack
- 6) Routing is responsibility of _____.
A. Presentation layer C. Physical layer
B. Network layer D. Session layer
- 7) A switch performs filtering on the basis of _____ address.
A. MAC C. port
B. logical D. postal
- 8) A Bluetooth Network is called a _____.
A. Picone C. Wireless
B. scatternet D. None of these
- 9) When the packets are small and all the same size, they are often called _____.
A. Cells C. Circuits
B. Trunks D. Subnet
- 10) _____ is an analog multiplexing technique that combines analog signals.
A. Multiplexing C. CDMA
B. Demultiplexing D. None of these

Q-2	Give short answers of the following questions: (Attempt TEN out of TWELVE)	20
1)	Explain DQDB in brief.	
2)	Explain SUBNET in brief.	
3)	Define terms : Topology, Data Rate.	
4)	What do you mean by Multiplexing? List out various Techniques.	
5)	What is a transmission media? List out various types (classes) of transmission media.	
6)	Write the difference between circuit switching and packet switching.	
7)	Define: Protocol Stack and Network architecture.	
8)	Explain orbit.	
9)	List main responsibilities of Physical Layer & Network layers.	
10)	Explain Bluetooth in short.	
11)	Explain function and list types of Hub.	
12)	Write full form of CSMA. List different types of CSMA.	
Q-3 A	Explain LAN in detail.	5
B	What are the uses/advantages & Disadvantages of computer networks?	5
OR		
Q-3 A	Explain WAN in detail.	5
B	Explain classification of computer Network.	5
Q-4 A	Explain twisted pair cable in detail.	6
B	Explain parallel transmission in detail.	4
OR		
Q-4 A	Explain FDM in detail.	6
B	Explain coaxial cable in detail.	4
Q-5 A	List all layers of OSI model. Explain any two layers.	6
B	Write a note on protocol hierarchies.	4
OR		
Q-5 A	Explain any two categories of satellite in detail.	5
B	Write a note on design issues for the layers.	5
Q-6	List and explain any two topologies in detail.	[10]
OR		
Q-6	Explain Repeater and Router in detail.	[10]